

Internet governance

Internet governance is the development and application of shared principles, norms, rules, decision-making procedures, and programs that shape the evolution and use of the Internet. No one person, company, organization or government runs the Internet. It is a globally distributed network comprising many voluntarily interconnected autonomous networks. It operates without a central governing body with each constituent network setting and enforcing its own policies. Its governance is conducted by a decentralized and international multistakeholder network of interconnected autonomous groups drawing from civil society, the private sector, governments, the academic and research communities and national and international organizations. They work cooperatively from their respective roles to create shared policies and standards that maintain the Internet's global interoperability for the public good.

Who-Runs-the-Internet-graphic

However, to help ensure interoperability, several key technical and policy aspects of the underlying core infrastructure and the principal namespaces are administered by the Internet Corporation for Assigned Names and Numbers (ICANN), which is headquartered in Los Angeles, California. ICANN oversees the assignment of globally unique identifiers on the Internet, including domain names, Internet protocol addresses, application port numbers in the transport protocols, and many other parameters. This seeks to create a globally unified namespace to ensure the global reach of the Internet. ICANN is governed by an international board of directors drawn from across the Internet's technical, business, academic, and other non-commercial communities. However, the National Telecommunications and Information Administration, an agency of the U.S. Department of Commerce, continues to have final approval over changes to the DNS root zone. This authority over the root zone file makes ICANN one of a few bodies with global, centralized influence over the otherwise distributed Internet. In the 30 September 2009 Affirmation of Commitments by the Department of Commerce and ICANN, the Department of Commerce finally affirmed that a "private coordinating process...is best able to flexibly meet the changing needs of the Internet and of Internet users". While ICANN itself interpreted this as a declaration of its independence, scholars still point out that this is not yet the case. Considering that the U.S. Department of Commerce can unilaterally terminate the Affirmation of Commitments with ICANN, the authority of DNS administration is likewise seen as revocable and derived from a single State, namely the United States.

The technical underpinning and standardization of the Internet's core protocols (IPv4 and IPv6) is an activity of the Internet Engineering Task Force (IETF), a non-profit organization of loosely affiliated international participants that anyone may associate with by contributing technical expertise.

On 16 November 2005, the United Nations-sponsored World Summit on the Information Society (WSIS), held in Tunis, established the Internet Governance Forum (IGF) to open an ongoing, non-binding conversation among multiple stakeholders about the future of Internet governance. Since WSIS, the term "Internet governance" has been broadened beyond narrow technical concerns to include a wider range of Internet-related policy issues. During 1979 the Internet Configuration Control Board was founded by DARPA to oversee the network's development. During 1984 it was renamed the Internet Advisory Board (IAB), and during 1986 it became the Internet Activities Board.

The Internet Engineering Task Force (IETF) was formed during 1986 by the U.S. government to develop and promote Internet standards. It consisted initially of researchers, but by the end of the year participation was available to anyone, and its business was performed largely by email.

From the early days of the network until his death during 1998, Jon Postel oversaw address allocation and other Internet protocol numbering and assignments in his capacity as Director of the Computer Networks

Division at the Information Sciences Institute of the University of Southern California, under a contract from the Department of Defense. This function eventually became known as the Internet Assigned Numbers Authority (IANA), and as it expanded to include management of the global Domain Name System (DNS) root servers, a small organization grew. Postel also served as RFC Editor.

Allocation of IP addresses was delegated to five Regional Internet Registries (RIRs):

1. American Registry for Internet Numbers (ARIN) for North America
2. Réseaux IP Européens - Network Coordination Centre (RIPE NCC) for Europe, the Middle East, and Central Asia
3. Asia-Pacific Network Information Centre (APNIC) for Asia and the Pacific region
4. Latin American and Caribbean Internet Addresses Registry (LACNIC) for Latin America and the Caribbean region
5. African Network Information Center (AfriNIC) was created in 2004 to manage allocations for Africa

After Jon Postel's death in 1998, IANA became part of ICANN, a California nonprofit established in September 1998 by the U.S. government and awarded a contract by the U.S. Department of Commerce. Initially two board members were elected by the Internet community at large, though this was changed by the rest of the board in 2002 in a poorly attended public meeting in Accra, Ghana.[24]

In 1992 the Internet Society (ISOC) was founded, with a mission to "assure the open development, evolution and use of the Internet for the benefit of all people throughout the world".[25] Its members include individuals (anyone may join) as well as corporations, organizations, governments, and universities. The IAB was renamed the Internet Architecture Board, and became part of ISOC. The Internet Engineering Task Force also became part of the ISOC. The IETF is overseen currently by the Internet Engineering Steering Group (IESG), and longer-term research is carried on by the Internet Research Task Force and overseen by the Internet Research Steering Group.

At the first World Summit on the Information Society in Geneva in 2003, the topic of Internet governance was discussed. ICANN's status as a private corporation under contract to the U.S. government created controversy among other governments, especially Brazil, China, South Africa, and some Arab states. Since no general agreement existed even on the definition of what comprised Internet governance, United Nations Secretary General Kofi Annan initiated a Working Group on Internet Governance (WGIG) to clarify the issues and report before the second part of the World Summit on the Information Society in Tunis 2005. After much controversial debate, during which the U.S. delegation refused to consider surrendering the U.S. control of the Root Zone file, participants agreed on a compromise to allow for wider international debate on the policy principles. They agreed to establish an Internet Governance Forum, to be convened by the United Nations Secretary General before the end of the second quarter of 2006. The Greek government volunteered to host the first such meeting.[26]

E-Government

E-government (short for electronic government, also known as e-gov, Internet government, digital government, online government, connected government) consists of the digital interactions between a citizen and their government (C2G), between governments and government agencies (G2G), between government and citizens (G2C), between government and employees (G2E), and between government and businesses/commerce (G2B).

Electronic government (or e-government) essentially refers to "The utilization of Information Technology (IT), Information and Communication Technologies (ICT s), and other web-based telecommunication technologies to improve and/or enhance on the efficiency and effectiveness of service delivery in the public sector." E-Government promotes and improves broad stakeholder's contribution to national and community development, as well as deepen the governance process.

E-government should enable anyone visiting a city website to communicate and interact with city employees via the Internet with graphical user interfaces (GUI), instant-messaging (IM), audio/video presentations, and in any way more sophisticated than a simple email letter to the address provided at the site" and "the use of technology to enhance the access to and delivery of government services to benefit citizens, business partners and employees". The focus should be on:

The use of information and communication technologies, and particularly the Internet, as a tool to achieve better government.

The use of information and communication technologies in all facets of the operations of a government organization.

The continuous optimization of service delivery, constituency participation and governance by transforming internal and external relationships through technology, the Internet and new media.

Whilst e-government has traditionally been understood as being centered on the operations of government, e-governance is understood to extend the scope by including citizen engagement and participation in governance. As such, following in line with the OECD definition of e-government, e-governance can be defined as the use of ICTs as a tool to achieve better governance.

Delivery models and activities of e-government

The primary delivery models of e-government can be divided into:

Government-to-citizen or government-to-consumer (G2C)

In this model, the G2C model apply the strategy of customer relationship management (CRM) with business concept.

By managing their customer (citizen) relationship, the business (government) can provide the needed products and services fulfill the needs from customer (citizen).

In United States, the NPR (National Partnership for Reinventing Government) has been implemented from 1993.

Government-to-business (G2B)

Government-to-government (G2G)

Government-to-employees (G2E)

Within each of these interaction domains, four kinds of activities take place:

1. Pushing information over the Internet, e.g.: regulatory services, general holidays, public hearing schedules, issue briefs, notifications, etc.
2. Two-way communications between the agency and the citizen, a business, or another government agency. In this model, users can engage in dialogue with agencies and post problems, comments, or requests to the agency.

3. Conducting transactions, e.g.: lodging tax returns, applying for services and grants.
4. governance, e.g.: To enable the citizen transition from passive information access to active citizen participation by:
 - a. Informing the citizen
 - b. Representing the citizen
 - c. Encouraging the citizen to vote
 - d. Consulting the citizen
 - e. Involving the citizen

Controversies of e-government

Disadvantages

The main disadvantages concerning e-government is the lack of equality in public access to the internet, reliability of information on the web, and hidden agendas of government groups that could influence and bias public opinions.

There are many considerations and potential implications of implementing and designing e-government, including disintermediation of the government and its citizens, impacts on economic, social, and political factors, vulnerability to cyber-attacks, and disturbances to the status quo in these areas

Hyper-surveillance

Increased contact between government and its citizens goes both ways. Once e-government begins to develop and become more sophisticated, citizens will be forced to interact electronically with the government on a larger scale. This could potentially lead to a lack of privacy for civilians as their government obtains more and more information on them. In a worst-case scenario, with so much information being passed electronically between government and civilians, a totalitarian-like system could develop. When the government has easy access to countless information on its citizens, personal privacy is lost.

Cost

Although "a prodigious amount of money has been spent" on the development and implementation of e-government, some say[who?] it has yielded only a mediocre product. The outcomes and effects of trial Internet-based governments are often difficult to gauge or unsatisfactory. According to Gartner, Worldwide IT spending is estimated to total \$3.6 trillion in 2011 which is 5.1% increase from the year 2010 (\$3.4 trillion).

Inaccessibility

An e-government site that provides web access and support often does not offer the "potential to reach many users including those who live in remote areas, are homebound, have low literacy levels, exist on poverty line incomes."

False sense of transparency and accountability

Opponents of e-government argue that online governmental transparency is dubious because it is maintained by the governments themselves. Information can be added or removed from the public eye. To this day, very few organizations monitor and provide accountability for these modifications. Those that do so, like the United States' OMBWatch[20] and Government Accountability Project, are often nonprofit

volunteers. Even the governments themselves do not always keep track of the information they insert and delete.

Advantages

The ultimate goal of the e-government is to be able to offer an increased portfolio of public services to citizens in an efficient and cost effective manner. E-government allows for government transparency. Government transparency is important because it allows the public to be informed about what the government is working on as well as the policies they are trying to implement. Simple tasks may be easier to perform through electronic government access. Many changes, such as marital status or address changes can be a long process and take a lot of paper work for citizens. E-government allows these tasks to be performed efficiently with more convenience to individuals. E-government is an easy way for the public to be more involved in political campaigns. It could increase voter awareness, which could lead to an increase in citizen participation in elections. It is convenient and cost-effective for businesses, and the public benefits by getting easy access to the most current information available without having to spend time, energy and money to get it.

E-government helps simplify processes and makes government information more easily accessible for public sector agencies and citizens. For example, the Indiana Bureau of Motor Vehicles simplified the process of certifying driver records to be admitted in county court proceedings. Indiana became the first state to allow government records to be digitally signed, legally certified and delivered electronically by using Electronic Postmark technology. In addition to its simplicity, e-democracy services can reduce costs. Alabama Department of Conservation & Natural Resources, Wal-Mart and NIC developed an online hunting and fishing license service utilizing an existing computer to automate the licensing process. More than 140,000 licenses were purchased at Wal-Mart stores during the first hunting season and the agency estimates it will save \$200,000 annually from service.

The anticipated benefits of e-government include efficiency, improved services, better accessibility of public services, sustainable community development and more transparency and accountability.

Democratization

One goal of e-government will be greater citizen participation. Through the internet, people from all over the country can interact with politicians or public servants and make their voices heard. Blogging and interactive surveys will allow politicians or public servants to see the views of the people they represent on any given issue. Chat rooms can place citizens in real-time contact with elected officials, their offices or provide them with the means to replace them by interacting directly with public servants, allowing voters to have a direct impact and influence in their government. These technologies can create a more transparent government, allowing voters to immediately see how and why their representation in the capital is voting the way they are. This helps voters better decide who to vote for in the future or how to help the public servants become more productive. A government could theoretically move more towards a true democracy with the proper application of e-government. Government transparency will give insight to the public on how decisions are made and hold elected officials or public servants accountable for their actions. The public could become a direct and prominent influence in government legislature to some degree.

Environmental bonuses

Proponents of e-government argue that online government services would lessen the need for hard copy forms. Due to recent pressures from environmentalist groups, the media, and the public, some governments and organizations have turned to the Internet to reduce this paper use. The United States government utilizes

the website <http://www.forms.gov> to provide “internal government forms for federal employees” and thus “produce significant savings in paper.

Speed, efficiency, and convenience

E-government allows citizens to interact with computers to achieve objectives at any time and any location, and eliminates the necessity for physical travel to government agents sitting behind desks and windows. Improved accounting and record keeping can be noted through computerization, and information and forms can be easily accessed, equaling quicker processing time. On the administrative side, access to help find or retrieve files and linked information can now be stored in databases versus hardcopies stored in various locations. Individuals with disabilities or conditions no longer have to be mobile to be active in government and can be in the comfort of their own homes.

Public approval

Recent trials of e-government have been met with acceptance and eagerness from the public. Citizens participate in online discussions of political issues with increasing frequency, and young people, who traditionally display minimal interest in government affairs, are drawn to electronic voting procedures.

Although internet-based governmental programs have been criticized for lack of reliable privacy policies, studies have shown that people value prosecution of offenders over personal confidentiality. Ninety percent of United States adults approve of Internet tracking systems of criminals, and 57% are willing to forgo some of their personal internet privacy if it leads to the prosecution of criminals or terrorists.

Technology-specific e-government

There are also some technology-specific sub-categories of e-government, such as m-government (mobile government), u-government (ubiquitous government), and g-government (GIS/GPS applications for e-government).

The primary delivery models of e-government are classified depending on who benefits. In the development of public sector or private sector portals and platforms, a system is created that benefits all constituents. Citizens needing to renew their vehicle registration have a convenient way to accomplish it while already engaged in meeting the regulatory inspection requirement. On behalf of a government partner, business provides what has traditionally, and solely, managed by government and can use this service to generate profit or attract new customers. Government agencies are relieved of the cost and complexity of having to process the transactions.

To develop these public sector portals or platforms, governments have the choice to internally develop and manage, outsource, or sign a self-funding contract. The self-funding model creates portals that pay for themselves through convenience fees for certain e-government transactions, known as self-funding portals.

Social networking services are an emerging area for e-democracy. The social networking entry point is within the citizens' environment and the engagement is on the citizens' terms. Proponents of e-government perceive government use of social networking as a medium to help government act more like the public it serves. Examples can be found at almost every state government portal through Facebook, Twitter, and YouTube widgets.